INTERNATIONAL JOURNAL OF
RADIATION ONCOLOGY • BIOLOGY • PHYSICS
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BRIEF REPORT

Tumor Control Probability and Time-Dose-Response Modeling for Stereotactic Radiosurgery of Uveal Melanoma

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Received Dec 13, 2023; Accepted for publication May 24, 2024

Purpose: Uveal melanoma (UM), although a rare malignancy, stands as the most prevalent intraocular malignancy in adults. Controversies persist regarding the dose dependency of local control (LC) through radiation therapy. This study sought to elucidate the significance of the prescription dose by employing time-dose-response models for patients with UM receiving photon-based stereotactic radiosurgery (SRS).

Methods and Materials: The analysis included patients with UM treated between 2005 and 2019. All patients underwent single-fraction SRS. Datapoints were separated into 3 dose groups, with Kaplan-Meier analysis performed on each group, from which time-dose-response models for LC were created at 2, 4, and 7 years after SRS using maximum-likelihood fitted logistic models.

Results: Outcomes from 594 patients with 594 UMs were used to create time-dose-response models. The prescribed doses and the number of patients were as follows: 17 to 19 Gy (24 patients), 20 Gy (122 patients), 21 Gy (442 patients), and 22 Gy (6 patients). Averaged over all patients and doses, LC rates at 2, 4, and 7 years were 94.4%, 88.2%, and 69.0%, respectively. Time-dose-response models for LC demonstrated a dose-dependent effect, showing 2-year LC rates of more than 90% with 20 Gy and 95% with 22 Gy. For 4 years and a LC of 90%, a dose of approximately 21 Gy was required. After 7 years, the 21 Gy prescription dose was predicted to maintain a LC above 70%, sharply declining to less than 60% LC with 19 Gy and less than 40% with 18 Gy.

Conclusions: In contrast to prior findings, the time-dose-response models for UM undergoing photon-based SRS emphasize the critical role of the prescription dose in achieving lasting LC. The dose selection must be carefully balanced against toxicity

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Disclosures: F.E. reports honoraria and travel support from ZAP Surgical Systems, Inc; C.F. reports honoraria from ZAP Surgical Systems, Inc, and Accuray, Inc, and is the current Vice Chair of the Physics Committee of The Radiosurgery Society; A.M. reports honoraria and travel support from ZAP Surgical Systems, Inc; J.G. reports grants from Accuray, Inc, and NovoCure, outside the submitted work; and has a patent DVH Evaluator issued.

Data Sharing Statement: Research data are not available at this time.

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Supplementary material associated with this article can be found in the online version at doi:10.1016/j.ijrobp.2024.05.025.

Int J Radiation Oncol Biol Phys, Vol. 120, No. 4, pp. 1052−1057, 2024

https://doi.org/10.1016/j.ijrobp.2024.05.025

risks, considering tumor geometry and individual patient characteristics to tailor treatments accordingly.

Introduction

Uveal melanoma (UM) is a rare malignant neoplasia that represents adults' most common intraocular malignancy.[1,2] Treatment options comprise brachytherapy (plaque radiation therapy), stereotactic radiosurgery (SRS), fractionated stereotactic radiation therapy, proton beam therapy, and enucleation.[1] Given its presumed radioresistance, radiation therapy for UM requires a considerable dose to achieve durable local control (LC), but controversies regarding dose dependency remain.[3-7] However, analyses of the time-dose-response and tumor control probability (TCP) are particularly limited in large cohorts but may help individualize treatment decision making in the management of patients. Patients are specifically interested in various time periods concerning their disease, such as survival, time of LC, time to next treatment, or the interval from treatment to anticipated toxicity.[8-11] Therefore, it is essential to incorporate the time to a specific outcome as a model of dose because managing physicians have the most direct control of dose. Given the lack of such analyses in the field of UM and remaining uncertainties concerning the impact of applied doses, this study reports the TCP and time-dose-response of UM receiving photon-based SRS.[4]

Methods and Materials

This analysis included patients with a confirmed UM diagnosis treated between 2005 and 2019 at our institution. Patients with prior treatments were excluded. A complete ophthalmologic examination was mandatory, including assessment of best-corrected visual acuity, slit-lamp biomicroscopy, funduscopy, ultrasound, and photographic documentation. Local tumor progression was also assessed with ophthalmologic assessments as well as imaging if deemed necessary. Treatment planning procedures and treatment delivery were executed as previously described.[12] In summary, all patients underwent single-fraction SRS using a robotic radiosurgery treatment platform (CyberKnife, Accuray Inc). Retrobulbar anesthesia was performed on the day of treatment. Then, contrast-enhanced magnetic resonance imaging was performed, acquiring T2-weighted and contrast-enhanced T1-weighted sequences. The following planning computed tomography was fused with the magnetic resonance imaging data. The results of the ophthalmologic examination informed the target delineation. An isotropic planning target volume margin of 1 mm was applied except for the posterior extension, which was set to 2 mm. Patients treated before 2012 were mostly treated with prescription doses of ≤20 Gy. The dose was uniformly changed to 21 Gy afterward. All treatments, including imaging acquisition and treatment planning, were delivered within 3 hours of retrobulbar anesthesia. Patients were staged according to the TNM (eighth edition) and American Joint Committee on Cancer (AJCC) classification system (I-IV). The AJCC staging incorporates the largest basal tumor diameter and thickness, information on ciliary body involvement, extrascleral extension, and presence of metastases. In general, tumor staging increases with tumor size and involvement of specific anatomic structures, eg, the ciliary body. Lymph node or other metastases indicate a stage IV disease. The overall prognosis gradually decreases with increased stage number. All patients had to have at least 1 available follow-up for analysis. To create the dose-response models, data from the available cases were partitioned into 3 dose groups, and Kaplan-Meier (KM) analysis was performed separately on each dose group. Then, time-dose-response models were created from each of the KM values, at each time point, ie, 2, 4, and 7 years, using maximum-likelihood fitted logistic models in the dose-volume histogram in the Evaluator software (DiversiLabs LLC).[13,14] The KM analysis was censored on length of follow-up and survival, and the resulting uncertainty at each dose-response modeling time point was assessed with 95% binomial confidence intervals (CIs), which were overlayed in the graphs as error bars. Time-to-event analysis was conducted using log-rank testing. Correlation assessment was conducted using Spearman's correlation coefficient. Analyses were done using STATA MP 17.0 (StataCorp LLC). *P* values below .05 were considered statistically significant. This study was approved by the local institutional review board.

Results

A total of 594 UMs in 594 patients were included in the analysis. The median age at the time of treatment was 64.5 years, ranging from 20.9 to 94.5 years. Most patients had AJCC stage II tumors (57.9%), followed by stages I (22.7%), III (18.9%), and IV (0.5%). Treatment decisions for stage IV patients were based on individual circumstances and patients' preferences. The median tumor height and largest basal diameter at the time of SRS were 5.8 mm (range, 0.7-16.0; standard deviation (SD), 3.10) and 11.4 mm (range, 1.2-22.0; SD, 3.5), respectively. The prescribed doses and respective number of patients were as follows: 17 to 19 Gy, 24 patients; 20 Gy, 122 patients; 21 Gy, 442 patients; and 22 Gy, 6 patients (Table 1). The target volume, ie, planning target volume, largest basal diameter, tumor height and prescription dose did show negligible relationships (all Spearman's rho > -0.20; all $P < .01$). The median prescription isodose line was 70%, and the median follow-up time was 35.1 months, ranging from 0.4 to 156.7

Table 1 Partitioning of the data for analysis

Raw data		PTV	Three dose levels for dose response		Two dose levels for log-rank testing	
Dose (Gy)	No. of cases	Median, range (cm^3)	Average dose (Gy)	No. of cases	Average dose (Gy)	No. of cases
17	1					
18	8	2.2 (0.43-3.60)	18.44	24	19.76	146
19	15					
20	122	1.37 (0.25-4.14)	20.00	122		
21	442	1.17 (0.18-4.29)	21.01	448	21.01	448
22	6	0.54 (0.28-3.61)				
Total	594	-	-	594	-	594

Abbreviation: PTV = planning target volume.

For dose-response modeling, the data were partitioned into 3 dose levels, whereas for comparison with Kaplan-Meier analysis, the data were also partitioned into 2 dose levels for hypothesis testing.

months. During the available follow-up, a total of 80 local failures were identified. Averaged over all patients and dose levels, the LC rates at 2, 4, and 7 years were 94.4% (95% CI, 92.3%-96.5%), 88.2% (95% CI, 84.9%-91.6%), and 69.0% (95% CI, 60.9%-75.8%), respectively. The calculated dose-response models for LC at 2, 4, and 7 years confirmed a dose-dependent effect, demonstrating 2-year LC rates of more than 90% with 20 Gy and more than 95% with 22 Gy (Fig. 1A). The probability of 2-year LC rapidly decreased with doses below 20 Gy. For 4 years and a LC of 90%, a dose of approximately 21 Gy was required, whereas for approximately 95% LC, a dose of more than 22 Gy was estimated (Fig. 1B). After 7 years, the 21 Gy prescription dose was predicted to have a LC above 70%, with a steep fall-off to less than 60% LC with 19 Gy and less than 40% with 18 Gy (Fig. 1C). Given the confined availability of 7-year follow-up data, the 95% CI was, however, markedly larger than that for the 2- and 4-year models (Fig. 1). With the data set partitioned into 2 dose groups, hypothesis testing demonstrated significance (Table 1, log-rank P =.03). The number of patients at risk at 2, 4, and 7 years was 379, 206, and 49, respectively (Figure E1). More than two-thirds of patients with local tumor recurrence underwent enucleation as salvage treatment.[15] The second most commonly used salvage treatment was reirradiation with SRS. The most frequent adverse radiation effect was serous retinal detachment, which was observed in 216 patients (36.3%). The detachment was already present at the time of SRS in 167 patients (28.1%). The second most frequent adverse radiation effect was secondary glaucoma (104 patients; 17.5%). Forty

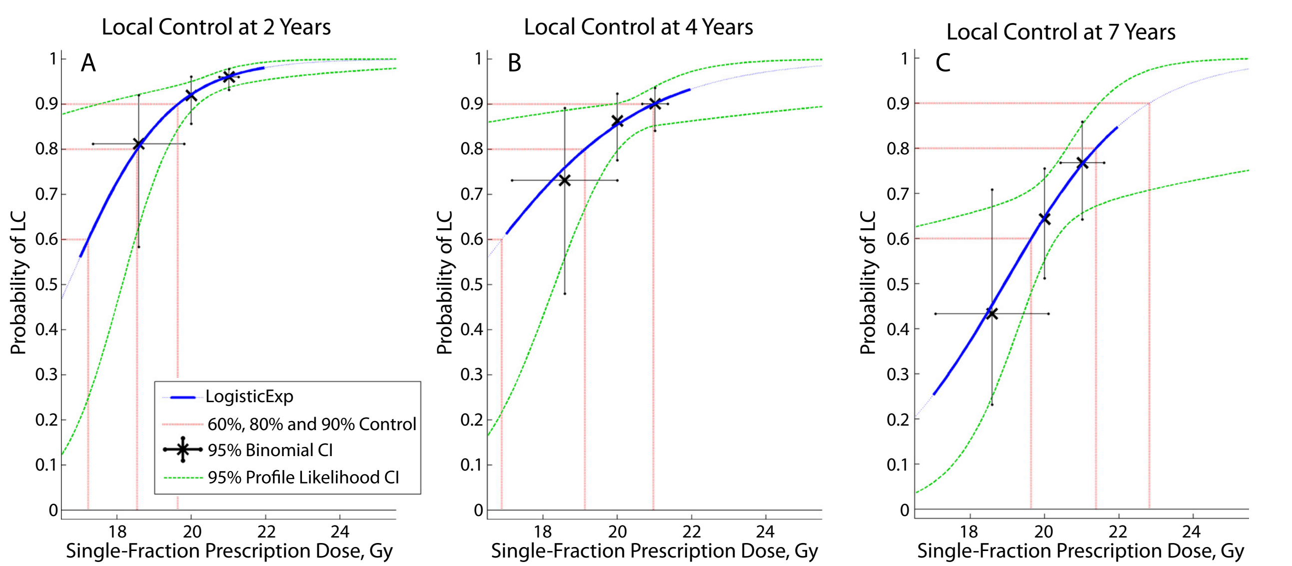

Fig. 1. Dose-response for uveal melanoma at (A) 2, (B) 4, and (C) 7 years after single-fraction stereotactic radiosurgery. The solid blue lines represent the maximum-likelihood fitted logistic model within the range of data, and the dashed blue lines show the predicted trend beyond currently available data. The dashed red lines indicate doses corresponding to 60%, 80%, and 90% local control (LC) according to the model. Profile likelihood 95% confidence intervals (CIs) are shown as dashed green lines.

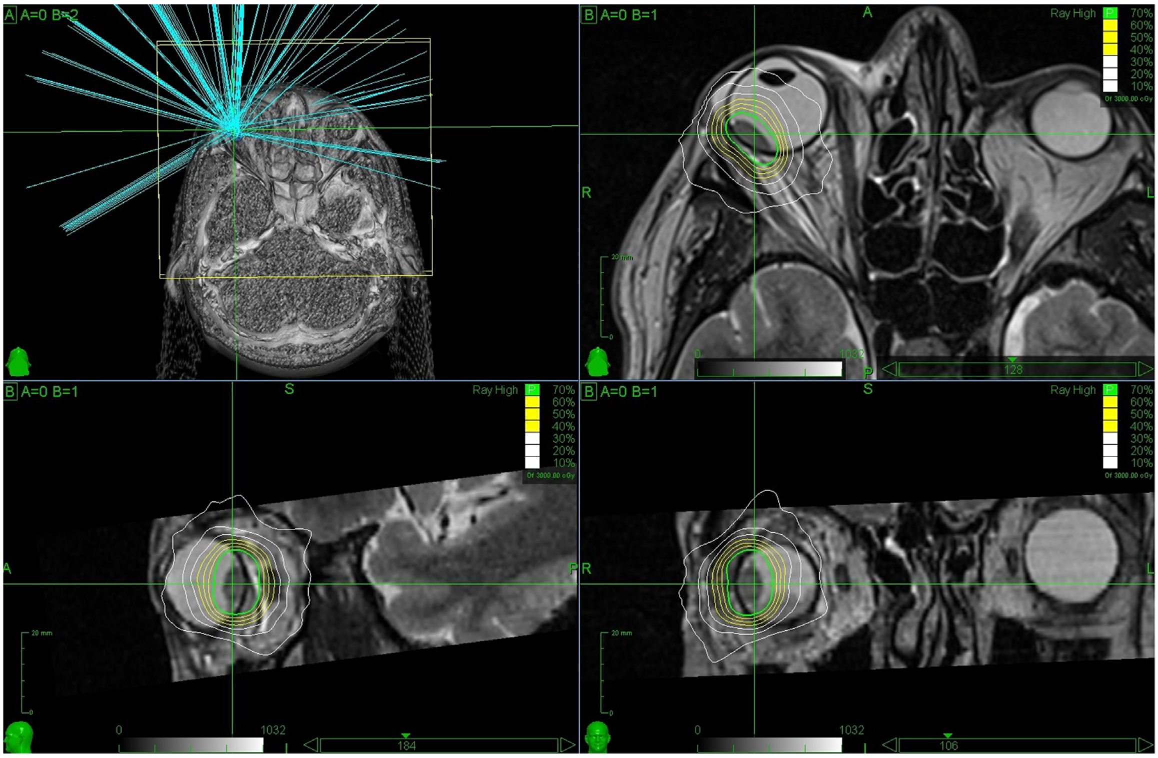

Fig. 2. Treatment plan of a 47-year-old patient with a right-sided uveal melanoma. Treatment was delivered with a prescription dose of 21 Gy and a prescription isodose line of 70%. Treatment planning was based on contrast-enhanced T1-weighted and T2-weighted magnetic resonance imaging as well as thin-slice computed tomography.

(38.4%) of the affected patients had to undergo surgical enucleation because of intractable secondary glaucoma. During the available follow-up, vitreous hemorrhage and radiation-induced retinopathy occurred in 74 (12.4%) and 72 patients (12.1%), respectively. A treatment case example is shown in Fig. 2.

Discussion

To our knowledge, this is the first study to report the TCP and time-dose-response models of a large patient cohort with UM receiving photon-based SRS. Given the central role of radiation therapy for UM, treatment planning and dose prescription remain 2 of the most crucial aspects in the management of patients.[1,4] To date, proton beam therapy, brachytherapy, fractionated stereotactic radiation therapy, and SRS seem to achieve comparable LC rates.[1,4,7,16] Although existing reports investigating these treatment modalities are mostly of retrospective nature, apply different definitions of local failure, and analyze heterogeneous patient cohorts that particularly differ in tumor staging and location, prospective randomized trials are lacking to determine the potential superiority of 1 modality. This ongoing uncertainty is also pertinent to the modality-related toxicity.[1,4,17] Although the toxicity profile of each treatment technique has specific nuances, it remains an active area of investigation to comprehensively characterize safety profiles.[17]

Moreover, controversies around the dose dependency of LC remain.[4-6,18] Previous studies have found an influence of the dose on LC, and others have not. The ongoing uncertainty is highlighted by the range of SRS doses recommended by the current National Comprehensive Cancer Network (18-45 Gy). It is important to note that previous analyses on this topic investigated not only photon-based teletherapy but also plaque brachytherapy, are limited by small sample sizes, and also analyzed fractionated stereotactic radiation therapy, highlighting the lack of in-depth studies in larger cohorts treated with photon-based SRS.[5,6,18] This is also particularly interesting because photon-based SRS is more accessible and cost effective than proton beam therapy, which is often considered a preferred treatment option.

Herein, our reported models for the LC of 594 patients at 2, 4, and 7 years confirm a dose-dependent effect (Fig. 1). A recent systematic review and meta-analysis on the use of Gamma Knife SRS for 898 UMs and uveal metastases with data on LC identified no dose dependency and highlighted the subsequent reduction of marginal doses over time (48/52 of reviewed studies reported data on the treatment of UM).[7] Our models contradict these findings, which might be caused by the inherent heterogeneity and size of analyzed studies in the systematic review, with the vast majority including sample sizes below 100 (17 of 19 studies included in the pooled analysis for local control; 89.4%).[7] Only 2 studies comprised more than 100 patients, 177 and 194, respectively.[7] The limitations that may arise from the distinct heterogeneities of these studies concerning sample sizes, treatment technique, prescription dose, isodose line, varying definitions of local treatment failure, tumor sizes, and staging can ultimately limit the possibility of detecting subtle effects such as dose dependency. At our institution, patients were consistently treated over the years in terms of treatment planning and delivery, including standardized follow-up in the Department of Ophthalmology. These circumstances reflect the possibility of identifying the time-dose-response as reported in our models. In light of our results, we would like to highlight that prescription dose and toxicity must be balanced and that dose selection must be carefully performed based on the tumor location, size, and other relevant factors. Yet, lower radiation doses (10-20 Gy), as underlined in the systematic review, and subsequent potential treatment failures can have devastating consequences for the patient.[7] Therefore, treatment of affected patients should be carried out at experienced centers with expertise in managing this rare tumor entity. Based on our findings, a clear recommendation for dose reduction cannot be made.

Modeling tumor control or normal tissue complication probabilities is of widespread interest in radiation oncology, but different approaches to include time dependencies exist. Herein, we chose to use maximum-likelihood fitted logistic models based on the KM estimates obtained.[13,14] Most recently, the effects of the time at which late complications occur compared with the length of follow-up and the survival time were modeled with the Cox proportional hazards technique in many of the Pediatric Normal Tissue Effects in the Clinic analyses.[19] The Cox proportional hazards method is an advanced technique with many advantages. However, it technically depends on the baseline hazard function, which was almost always omitted in the Pediatric Normal Tissue Effects in the Clinic source data sets. Alternatively, our proposed model can be used with any KM estimates and data sets that have been stratified into sufficiently small dose or volume parts. The advantage of using actuarial estimates is that they show the response after treatment as a function of time, and given that managing physicians have primary control over dose, we combined dose response with KM actuarial estimates.

The limitations of this work, such as the retrospective nature of the analysis and the potential for sampling biases, should be acknowledged. The diverse patient population and the inclusion of varying tumor stages enhance the generalizability of the findings but may also introduce heterogeneity. Treatment-associated toxicity is also of considerable importance for patients. Analyses of them and their dose dependency are crucial but out of the scope of the current study.

In conclusion, this study significantly contributes to the understanding of the time-dose-response and TCP after photon-based SRS for UM. The dose-dependent effect demonstrated by our models underscores the critical role of radiation dose in achieving and sustaining LC for this rare tumor entity. The findings provide valuable insights for clinicians managing UM, aiding in treatment decision making and also highlighting the necessity for long-term surveillance to capture the nature of tumor response after treatment. The reported time-dose response models need to be validated in the future and can act as a source to inform future research in the field.

Conclusions

The reported time-dose response models of UMs receiving SRS underline the importance of the prescription dose to obtain a durable LC. The choice of prescription dose should be weighed against the risk of toxicity, given the tumor geometry and characteristics of each patient, to individualize treatments. Although KM analysis depicts the response as a function of time, managing physicians have control over the dose, so the most informed treatment decision making needs dose-response models, including the time kinetics. To overcome the limitations of retrospective data with known and unknown confounders, future refinements with propensity-matched analyses and randomized controlled trials are proposed to validate our reported time-dose models and the role of dose dependency for LC in UMs.

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